NOTES ON MEASURE THEORY BY COHN

Quinten Grubbs
University of Missouri-St. Louis
December 15 2025

Contents

1 Measures **3**

1.1 Algebras and Sigma-Algebras 3

1 Measures

1.1 Algebras and Sigma-Algebras

Definition: Let X be an arbitrary set. A collection $\mathcal{A}$ of subsets of X is an *algebra* on X if

1. $X \in \mathcal{A}$
2. for each set A that belongs to $\mathcal{A}$, the set $A^c \in \mathcal{A}$
3. for each finite sequence $A_1, ..., A_n$ of sets that belong to $\mathcal{A}$, the set $\cup_{i=1}^{n} A_i$ belongs to $\mathcal{A}$
4. for each finite sequence $A_1, ..., A_n$ of sets that belong to $\mathcal{A}$, the set $\cap_{i=1}^{n} A_i$ belongs to $\mathcal{A}$

We now define something similar called a σ-algebra if:

1. $X \in \mathcal{A}$
2. for each set A that belongs to $\mathcal{A}$, the set A^c belongs to $\mathcal{A}$
3. for each infinite sequence $\{A_i\}$ of sets that belong to $\mathcal{A}$, the set $\cup_{i=1}^{\infty} A_i$ belongs to $\mathcal{A}$
4. for each infinite sequence $\{A_i\}$ of sets that belong to $\mathcal{A}$, the set $\cap_{i=1}^{\infty} A_i$ belongs to $\mathcal{A}$

- If X is a set and $\mathcal{A}$ is a family of subsets of X that is closed under complementation, then X belongs to $\mathcal{A}$ iff $\emptyset \in \mathcal{A}$
- If $\mathcal{A}$ is a family of subsets of X that is nonempty, closed under complementation, and closed under the formation of finite or countable unions, then $\mathcal{A}$ must contain X.

Some examples of σ-algebras are as follows:

- Let X be a set, then the powerset of X is a σ-algebra.
- Let X be a set, and let $\mathcal{A} = \{\emptyset, X\}$. Then $\mathcal{A}$ is a σ-algebra
- Let X be an infinite set, and let $\mathcal{A}$ be the collection of all finite subsets of X. Then $\mathcal{A}$ doees not contain X and is not closed under complementation; hence it is not an algebra (or a σ-algebra on X.

Proposition: Let X be a set. Then the intersection of an arbitrary nonempty collection of σ-algebras on X is a σ-algebra on X.
Proof: Let $\mathcal{C}$ be a nonempty collection of σ-algebras on X. Now, let $\mathcal{A}$ be the intersection of the σ-algebras that belong to $\mathcal{C}$. We just need to check that $\mathcal{A}$ contains X, is closed under complementation, and is closed under the formation of countable unions. The set X belongs to $\mathcal{A}$ since each σ contains X. Now, let $A \in \mathcal{A}$, each σ-algebra that belongs to $\mathcal{C}$ contains A and so also contains $\mathcal{A}^c$. Finally, we suppose $\{A_i\}$ is a sequence of sets that belong to $\mathcal{A}$ and hence to each σ-algebra in $\mathcal{C}$. Then $\cup_i A_i$ belongs to each σ-algebra in $\mathcal{C}$ and so to $\mathcal{A}$.

This proposition then implies the following result which is useful for the construction of σ-algebras:
Corollary: Let X be a set, and let $\mathcal{F}$ be a family of subsets of X. Then there is a smallest σ-algebra on X that includes $\mathcal{F}$.
This smallest σ-algebra is, as it turns out, unique. The smallest σ-algebra is called the σ-algebra generated by $\mathcal{F}$ and is denoted by $\sigma(\mathcal{F})$.

Proof: Let $\mathcal{C}$ be the collection of all σ-algebras on X that include $\mathcal{F}$. Then $\mathcal{C}$ is nonempty, since it contains the σ-algebra that consists of all subsets of X. The intersection of the σ-algebras that belong to $\mathcal{C}$ is, according to the previous proposition, a $\sigma - algebra$; it includes $\mathcal{F}$ and is included in every σ-algebra in $\mathcal{C}$- that is, it is included in every σ-algebra in X that includes $\mathcal{F}$.

The *Borel σ-algebra* on $\mathbb{R}^d$ is the $\sigma - algebra$ on $\mathbb{R}^d$ generated by the collection of open subsets of $\mathbb{R}^d$; it is denoted by $\mathcal{B}(\mathbb{R}^d$.